

Important Links

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Strategy + Vision

The Story

I am Sanskar Gupta. I breezed through freshman year of high school without ever really studying — not because I was lazy, but because I never had to. A 20-minute review before a test was enough. Nobody told me that was a problem until it became one.

Sophomore year arrived with AP Chemistry, AP US History, AP Psychology. Classes that don't reward raw intelligence or last-minute cramming. They reward consistency, depth, and genuine understanding built over weeks. For the first time in my life, I hit a wall. I sat down to study and genuinely didn't know what to do. Reading notes felt passive and pointless. Watching YouTube videos felt productive until I closed the tab and remembered nothing. Flashcard apps felt mechanical. Nothing had a feedback loop. Nothing told me whether what I was doing was actually working until the test came back with a number on it.

I looked for a tool that could help. What I found were tools designed for people who already knew how to study. Apps that assumed you'd show up, do the work, and trust the process — none of them engineered around how memory actually works. None of them knew anything about my material, my deadlines, or which concepts I was genuinely confused about versus which ones just needed one more pass. I was a first-generation student in the American school system. My parents couldn't help. There was no tutor, no older sibling who'd been through AP classes, no college counselor pointing me toward the right resources. I was navigating a system that wasn't designed for people like me, using tools that weren't built for my situation.

So I built Veridian.

What Veridian Is

Veridian is a personalized learning engine that turns your own study material into an active, adaptive practice system — one that knows what you don't know yet, drills you on it until you do, and tells you exactly where you stand before the exam, not after.

The core belief behind it is simple: most students don't fail because they aren't smart or don't work hard. They fail because they study passively, study the wrong things, and have no feedback loop between effort and outcome until it's too late to change anything. Veridian closes that loop.

A student uploads their class notes, a textbook chapter, a study guide, or just describes a topic. Veridian ingests the material, extracts the concepts and knowledge structure inside it, and builds a personalized learning Journey. That Journey has three stages: first exposure and guided understanding, then unlimited AI-generated practice that never repeats the same question twice, then mastery verification through active recall and synthesis. The entire system is scheduled by FSRs — the most accurate spaced repetition algorithm known — calibrated to the student's actual exam deadline so that review intervals are optimized to hit mastery at exactly the right moment.

The intelligence behind Veridian is powered by Gemini Flash — fast, cheap, and capable of handling any subject a student uploads. For users who want complete privacy, a local WebGPU mode processes everything on-device with nothing leaving the browser. Veridian is and will always be completely free. Not as a launch promotion — as a permanent commitment. The people who need this most are often the ones who can least afford another subscription.

The Problem, Stated Honestly

Most students don't fail because they aren't capable. They fail because:

- **They study the wrong things** — time spent on what they already know instead of what they're about to forget
- **They study in the wrong way** — passive re-reading and highlighting instead of active retrieval and testing
- **They study at the wrong times** — cramming the night before instead of spacing review over weeks
- **Their study sessions are disconnected from their actual material** — generic flashcard apps and YouTube videos are not calibrated to the specific content on their specific exam
- **They have no feedback loop** — no way to know whether what they're doing is working until the grade comes back

Every one of these failures is solvable with known learning science. Spaced repetition solves the timing problem. Active recall solves the method problem. Source-grounded AI question generation solves the material alignment problem. Knowledge gap detection solves the feedback problem. Veridian implements all four, integrated into a single system that requires no prior knowledge of how to study effectively.

Who This Is For

Veridian is built for the student who is capable and knows it — but whose environment hasn't given them the tools or the roadmap to perform at their level.

Specifically, the first version of Veridian is built for **AP students**. This is the deliberate wedge and it is chosen carefully, not arbitrarily.

AP students are the perfect first community for three reasons. First, the pain is acute and measurable — AP exams have fixed dates, fixed formats, and a 1-5 score that is an undeniable outcome metric. When a student's score improves, it is visible and attributable. Second, the material is standardized enough that Veridian's AI can generate high-quality practice questions from any uploaded AP material without hallucinating irrelevant content — AP Chemistry has a defined scope; Gemini knows it well. Third, the community is self-organized and evangelical. AP students talk to each other on Reddit's r/APStudents, on Discord servers, in their schools. A student who uses Veridian for two weeks before an AP exam and jumps from a 3 to a 5 will tell every person they know. That word-of-mouth is the growth engine, not advertising.

Beyond this wedge, Veridian is built for anyone navigating a system that wasn't designed for them — the first-generation student without college counselors or tutoring, the self-learner without a classroom, the student who is smart enough to know their current approach isn't working but doesn't know what to change. The product serves all of them. The story starts with AP students because those communities can be reached, measured, and mobilized first.

What Sets Veridian Apart

Against NotebookLM

Google's NotebookLM uploads your material and lets you interact with it conversationally. It is a comprehension and research tool. It does not have spaced repetition, does not track what you've practiced, does not know your exam date, does not detect which concepts you keep getting wrong and surface them more aggressively, and does not generate novel questions that never repeat. It is a search interface for your notes. Veridian is a training system built from them. These are genuinely different products.

Against Anki

Anki is the gold standard of spaced repetition and has been for 15 years. It is also, genuinely, a terrible experience for most people — dated interface, painful setup, zero AI, and it does exactly one thing. Veridian uses the same underlying algorithm (FSRS, the most accurate spaced repetition model available) but wraps it in an experience that requires no configuration, builds practice material from the student's own uploaded content automatically, and layers AI question generation on top so the student is never practicing the same question twice.

Against Quizlet

Quizlet is where students go when a teacher makes them. It is optimized for making flashcard sets quickly, not for learning deeply. Its algorithm is not scientifically rigorous and its questions are gameable — students memorize the card, not the concept. Veridian generates novel questions every session from the same knowledge base, making pattern memorization impossible. The student is always being tested on understanding, not recognition.

Against Khan Academy

Khan Academy provides content — videos, exercises, a structured curriculum. It is a content platform built around Khan Academy's material. Veridian is a learning intelligence layer built around the student's own material. These are not competing products. A student can watch Khan Academy's video on a topic and then come to Veridian to be tested on it using their class notes. They serve adjacent needs. The meaningful difference is that Veridian knows your specific textbook, your specific teacher's slides, your specific exam date — Khan Academy knows none of these things.

The real moat

Veridian's defensibility is not any single feature. It is the combination of FSRS-calibrated scheduling, source-material grounding, novel AI question generation, and knowledge gap detection — and the user data that accumulates over time. The longer a student uses Veridian, the more the system knows about their specific knowledge gaps and learning patterns. That personalization compounds. A student who has used Veridian for six months has a system that is genuinely tailored to them in a way that cannot be replicated by a competitor starting from scratch. That is the moat — accumulated learning intelligence, not just a feature list.

The Strategic Pillars

1. Science first, always.

Every feature Veridian builds that touches learning has a scientific basis. FSRS is not a random shuffling algorithm — it is a mathematical model of human memory developed from decades of cognitive science research. Active recall is not a gimmick — it is one of the most replicated findings in educational psychology, producing retention rates 50% higher than passive re-reading. Novel question generation prevents the fluency illusion — the false sense of knowing that comes from seeing the same question repeatedly. When Veridian tells a student to review something today and not tomorrow, there is a reason grounded in research. The system earns trust by being right.

2. Your material, not ours.

Veridian is a learning intelligence layer, not a content platform. The student brings their textbook, their notes, their study guide — material that is already aligned to their specific class, their specific teacher, their specific exam. Veridian extracts the knowledge inside that material and builds the practice system around it. This is not a replication of what Khan Academy or Coursera does. It is the thing they cannot do — personalization to the student's actual academic situation rather than a generic curriculum.

3. Free forever, no exceptions.

Not as a marketing claim. As a permanent structural commitment. Every feature Veridian has is available to every student who opens it. There will never be a premium tier, a Pro plan, or a feature locked behind a paywall. The students who need this most — first-generation, low-income, under-resourced — are often the ones who can least afford another subscription. Sal Khan built Khan Academy on this principle and it became the most democratizing educational tool in history. Veridian holds the same line.

4. Honest feedback over comfortable feedback.

Veridian will not tell a student they're doing well when they're not. If a student has 40 concepts below mastery threshold and an exam in four days, the system says that clearly and tells them exactly what to prioritize. Encouragement without accuracy is just noise. The goal is not to make studying feel good — it is to make studying actually work.

5. Build for the community, grow through it.

Veridian's growth mechanism is native to the product. When a student finishes a Journey on AP Chemistry Unit 4, they can publish it to the Community Library. Other AP Chemistry students find it, clone it, adjust it to their deadline, and use it. Each student who gets a 5 on their exam because of a community Journey tells their friends. Each friend who joins and creates a Journey adds to the library. The community builds the content. Veridian provides the intelligence layer on top of it. This flywheel — content in,

outcomes out, community growth through results — is the organic distribution engine. It is not a social network. It is a knowledge commons.

What Veridian Is Not

Veridian is not a social platform. There is no feed, no following, no public profiles, no algorithmic pressure to post or perform. The social layer exists for one purpose: to let students share what's working so other students can benefit from it.

Veridian is not a note-taking app. It is not trying to replace where students write their class notes. It turns those notes into something active and testable — not a record that sits unread until the night before the exam.

Veridian is not a content platform. It does not write the curriculum. The student brings their material. Veridian builds the training system from it.

Veridian is not trying to make studying feel effortless. Genuine learning is effortful by design — the difficulty is what makes it stick. What Veridian removes is wasted effort: time spent reviewing things already known, passive re-reading that produces no retention, studying without knowing if it's working. What remains is productive struggle, precisely directed.

The Vision

Two years from now, Veridian is the tool that AP students reach for first — not because it was marketed to them, but because someone in their class or study group said "this is what I used" and the result spoke for itself. It is the Community Library that has become the standard resource for AP exam prep, built not by a content team but by students who used it and wanted to give back.

Five years from now, Veridian is the study system for the student who doesn't have tutors, college counselors, or parents who went to American universities. It is the tool that closes the gap between the student with every advantage and the student with none — not by lowering the bar but by raising the floor of what serious, effective studying looks like when you have the right system.

The long-term vision is unchanged from the day it was conceived: genuinely effective learning, available to every student who needs it, free forever, grounded in science, and powerful enough to actually change outcomes.

Not the most popular study app. The best one. And free for everyone who needs it.

PRD

Landing Page

App Home

Overview

The App Home is the logged-in landing screen. Every time a student opens Veridian, this is where they arrive. Its single job is to answer one question immediately: **what should I do right now?** It is not a dashboard for reflection, not a stats overview, and not a feature showcase. It is an action surface — designed to get the student from opening the app to actively studying in as few seconds as possible.

Navigation Structure

Desktop — Left Sidebar

A persistent left sidebar runs the full height of the screen. It contains four navigation items, represented as icons with labels:

- **Home** — the App Home screen
- **Journeys** — the student's full Journey library
- **Community** — the Community Journey Library
- **Profile** — account settings, personal analytics, notification preferences, and privacy controls (settings live here, no separate settings page)

At no point does the sidebar contain content, alerts, or dynamic data. It is purely navigational. The active item is visually indicated with a filled icon or accent highlight.

The top of the sidebar displays the **Veridian wordmark**. Clicking it from anywhere in the app returns the student to Home. There is no other persistent header element.

Mobile — Bottom Tab Bar

On mobile, the sidebar collapses entirely and is replaced by a bottom tab bar with the same four items as icons. The Veridian wordmark moves to a slim top bar on mobile, still tappable to return Home. No hamburger menus, no hidden navigation.

Main Area Layout

The main content area of the App Home is organized into two vertical zones:

1. **The Due Today Zone** — at the top, showing all Journeys with activity due today

2. **The Journey Grid** — below, showing all active Journeys with a New Journey entry point

These two zones are always visible on a single scroll. The page is not paginated and does not require the student to navigate anywhere to see their full picture for the day.

Zone 1 — Due Today

Purpose

This zone surfaces every Journey that has something due today — FSRS-scheduled card reviews, a recommended Stage B practice session triggered by deadline proximity, or a Stage A concept the system flagged as incomplete. It is the primary action driver on the home screen.

Layout

Due Today items appear as a **horizontal scrollable row of action cards** if there are multiple, or a single full-width card if only one item is due. Each card represents one actionable unit from one Journey.

What Each Due Today Card Shows

- **Journey name and subject** — e.g., "AP Chemistry — Unit 4"
- **Action type** — clearly labeled: "12 cards due for review," "Practice session recommended," "Complete Stage A — Equilibrium"
- **Urgency context** — days until the Journey's exam deadline, displayed as a countdown: "Exam in 6 days"
- **Estimated time** — approximate session length: "~15 min"
- **A single launch button** — tapping it goes directly into that activity with no intermediate screen

The cards are ordered by urgency. A Journey with an exam in 3 days appears before one with an exam in 14 days, regardless of how many cards are due in each. If two Journeys have equal deadline proximity, the one with more cards due appears first.

The Caught Up State

When nothing is due today — all reviews are scheduled for future dates and no deadlines are critically close — the Due Today zone displays a **caught up card**. It

communicates this clearly without making it feel like the app has nothing to offer. The caught up card shows:

- A clear message: "You're all caught up for today."
- Three suggested next actions presented as tappable chips:
 - **Explore Community** — links to the Community Journey Library
 - **Start a New Journey** — opens the Journey creation flow
 - **Mastery Check** — prompts the student to pick a completed Journey and run a Stage C mastery session on it

These three suggestions are always the same in the caught up state. They are not personalized or rotated — they represent the three genuinely useful things a student can do when their scheduled work is done, and familiarity with them is a feature not a limitation.

Zone 2 — Your Journeys

Purpose

This zone gives the student a visual overview of all their active Journeys and serves as the primary entry point for creating new ones. It is not the full Journey management page — that lives in the Journeys section of the sidebar. This zone is a contextual at-a-glance view of what the student is currently working on.

Header

A section label — "**Your Journeys**" — sits at the top left of this zone. On the right side of the same row is a "**+ New Journey**" button. This means a student can start a new Journey directly from the home screen without navigating away. The New Journey button opens the Journey creation flow as an overlay or sheet, not a full page navigation.

Journey Cards Grid

Active Journeys are displayed in a **responsive card grid** — two columns on desktop, one column on mobile. Cards are ordered by deadline proximity, closest first.

Each Journey card displays:

- **Subject and title** — e.g., "AP Chemistry" as the primary label, with the Journey name below it if the student named it something custom
- **Current stage** — a subtle pill indicator showing which stage the student is currently in: Stage A, Stage B, or Stage C
- **Deadline countdown** — days remaining until the exam date: "Exam in 14 days." If no deadline was set, this shows "No deadline set" in muted text
- **Progress indicator** — a thin progress bar or ring showing rough concept mastery percentage, derived from FSRs card stability scores across the Journey's knowledge map. This is not a percentage of cards completed — it reflects actual retention quality
- **Last studied** — a small timestamp: "Last studied 2 days ago." If a Journey has not been touched in over 5 days and has an exam within 2 weeks, this timestamp is displayed in a warning color to signal neglect without being alarmist

Clicking anywhere on a Journey card opens the Journey Detail page for that Journey.

Archived Journeys

Active Journeys are the default view. A clearly labeled **"Archived"** tab appears at the top of the Journey Grid zone, next to an implied "Active" tab that is selected by default. Switching to Archived shows all past Journeys — completed, manually archived, or past their exam date — in the same card format but with muted styling to indicate they are not current. Students can un-archive a Journey from this view.

There is no separate page for archived Journeys. The tab system keeps both states accessible from the home screen without requiring navigation to the full Journeys section.

Empty State — New User

When a student logs in for the first time and has no Journeys, the main area shows a purpose-built empty state instead of both zones described above. It contains:

- A brief orienting statement — one or two sentences that describe what Veridian does and what the student should do first
- Two primary action buttons: **"Create Your First Journey"** and **"Browse Community Journeys"**
- A small example of what a Journey card looks like, rendered as a visual preview with placeholder data, so the student understands what they are working toward

The empty state is never shown again after the student creates their first Journey. It exists solely to reduce the blank canvas problem for a first-time user.

Notification Badge

Veridian surfaces a notification badge on the Home icon in the sidebar whenever reviews are due or an exam deadline is approaching. Notification preferences — email digests, urgent-only alerts, or off entirely — are configured in Profile and do not affect the Home screen itself beyond this badge.

Journeys

Overview

A Journey is the top-level container for everything a student is working toward in Veridian. It represents a single learning objective with a defined scope, a deadline, and a structured path from first exposure to mastery. Every study activity in Veridian lives inside a Journey.

The Journeys page is the student's full Journey library — the canonical management view for everything they have created. Unlike the Home screen, which is action-driven, the Journeys page is organizational. Students come here to create new Journeys, review what they are working on, manage existing ones, and access archived work.

The Journey Model

A Journey is organized into three layers:

Journey — the container. Holds the topic, deadline, source material, and overall schedule.

Modules — the conceptual units inside a Journey. Each Module represents a distinct topic or concept extracted from the student's source material. Modules are the organizing structure of the Journey and each one progresses through stages independently based on the student's demonstrated knowledge in that area.

Activities — the actual study sessions inside each Module. Each activity is one of Veridian's study modes: a Learning Guide (Stage A), a Practice Quiz session (Stage B), a Flashcard set, or a Mastery Check (Stage C). Activities are generated automatically by Veridian's AI from the source material but can be manually edited, supplemented, or added to by the student at any time.

Students never interact with the word "deck." The mental model is always Journey → Modules → Activities. The underlying FSRS scheduling and card data is infrastructure — it drives the system invisibly and surfaces to the student only as recommendations and progress indicators.

Journeys Page Layout

The Journeys page has two tabs at the top of the content area: **Active** (default) and **Archived**. Below the tabs, a **" + New Journey"** button sits in the top right of the content area.

Active Tab

Displays all current Journeys as a responsive card grid — two columns on desktop, one column on mobile. Cards are ordered by deadline proximity, closest deadline first. Each Journey card shows:

- Subject and Journey name
- Number of modules and total activities
- Current overall stage distribution — e.g., "3 modules in Stage B, 1 in Stage C"
- Deadline countdown — "Exam in 9 days"
- Overall mastery progress bar derived from FSRS stability scores across all modules
- Last studied timestamp, highlighted in warning color if more than 5 days have passed with an exam within 2 weeks

Clicking a Journey card opens the Journey Detail page for that Journey.

Archived Tab

Displays all past Journeys — completed, manually archived, or past their exam date — in the same card format with muted styling. Students can un-archive any Journey from this view, which returns it to Active with its full history and FSRS data intact. Archived Journeys are never deleted automatically.

Creating a Journey

Creating a Journey is a multi-step flow triggered by clicking "+ New Journey" from either the Home screen or the Journeys page. The flow is designed to be fast — a student should be able to go from clicking the button to having a fully structured Journey ready to study in under 5 minutes.

Step 1 — Basic Setup

The student provides three inputs:

- **Subject or topic name** — free text, e.g., "AP Chemistry" or "Organic Chemistry — Semester 1"

- **Exam or target date** — a date picker. This is the deadline the entire FSRS schedule is calibrated around. It is required, not optional, because without it the system cannot schedule reviews intelligently. If the student genuinely has no deadline, they select "No deadline" which switches the system to an open-ended mastery mode rather than deadline-calibrated scheduling
- **Prior knowledge level** — a three-option selector: "Starting from scratch," "I know some of this," or "I know most of this." This is a coarse self-assessment that influences the diagnostic and initial module placement. It is not a replacement for the diagnostic — it is a signal that helps calibrate how the diagnostic is structured

Step 2 — Source Material

The student uploads or provides the material Veridian will use to build the Journey. Multiple input methods are supported:

- **PDF upload** — textbook chapters, class notes, study guides, past exams
- **Text paste** — raw text pasted directly into a text area
- **Google Docs link** — the student pastes a shareable link and Veridian fetches the content
- **YouTube URL** — Veridian extracts the transcript and uses it as source material
- **Photo or image upload** — OCR processing extracts text from handwritten or printed notes
- **Topic only** — the student provides just a topic name (e.g., "AP Chemistry Unit 4: Chemical Reactions") and Veridian generates the Journey from its own knowledge of that subject without student-provided material. This is the lowest-friction option and is appropriate for well-defined standardized subjects

Multiple sources can be combined. A student can upload a PDF and paste additional notes and add a YouTube lecture URL, and Veridian will synthesize all three into a unified knowledge base for the Journey.

Step 3 — Module Proposal

After processing the source material, Veridian's AI analyzes the content and proposes a set of Modules — the distinct conceptual units it identified. This is presented as a structured review screen:

- The proposed modules are listed with a short one-sentence description of what each covers

- The student can **approve as-is**, or make changes: rename a module, merge two modules that overlap, delete a module that isn't relevant to their exam, or add a custom module not detected in the material
- A module count is shown at the top — "Veridian found 6 concepts in your material." If the student's source material is very broad (e.g., an entire AP class), Veridian may propose more modules and note that the Journey will be larger and longer. If the material is narrow (a single chapter), it may propose 3-4 focused modules
- The student confirms the final module list with a single button

This step is intentionally lightweight. It is not a configuration wizard — it is a quick sanity check that takes under a minute. The default proposal is almost always usable as-is for a student who doesn't want to customize anything.

Step 4 — The Diagnostic

Once modules are confirmed, Veridian presents the diagnostic. This is framed explicitly as a low-stakes knowledge check, not a test:

"Let's see what you already know — 15 quick questions across your modules. No grades, no pressure. This takes about 5 minutes and means Veridian won't make you re-learn things you already know."

The diagnostic consists of 15-20 AI-generated questions distributed across all confirmed modules, weighted so each module gets at least one question and larger modules get proportionally more. Questions are multiple choice and conceptual — designed to probe genuine understanding rather than surface-level recall.

The student can skip the diagnostic with a clearly visible but secondary "Skip — start everything from Stage A" option. The benefit of completing it is stated plainly next to the skip option so that skipping feels like leaving value on the table rather than a neutral choice.

Diagnostic results are processed immediately. The student sees a brief summary — "You're solid on Stoichiometry and Thermodynamics. Equilibrium and Electrochemistry need work." — and then each module is automatically assigned a starting stage:

- **Stage A** — little to no existing knowledge demonstrated. The module begins with a structured Learning Guide before any practice

- **Stage B** — partial knowledge demonstrated. The module skips directly to AI-generated practice, using gaps identified in the diagnostic to prioritize which concepts to hit first
- **Stage C** — strong existing knowledge demonstrated. The module goes straight to Mastery Check activities to confirm and formalize what the student already knows

If the diagnostic is skipped, all modules default to Stage A regardless of the prior knowledge selection in Step 1. A banner on the Journey Detail page notes that the diagnostic was skipped and offers the option to run it at any time.

Step 5 — Journey Created

The Journey is created and the student lands on the Journey Detail page. Veridian displays a clear first action — the highest-priority activity based on deadline proximity and module stage assignments. The student can begin immediately or return to it later from Home.

Journey Detail Page

The Journey Detail page is the full view for a single Journey. It is the operational hub for everything related to that Journey — progress, modules, activities, settings, and sharing.

Header

- Journey name and subject
- Deadline countdown — large and prominent
- Overall mastery progress — a single percentage derived from FSRS stability across all modules, with a progress bar
- Edit button — opens inline editing for Journey name, deadline, and privacy setting
- Share button — described in the Sharing section below

Module Overview

The central element of the Journey Detail page is the **Module Overview** — a list or grid of all modules in the Journey, each showing:

- Module name

- Current stage (A, B, or C) displayed as a colored pill
- Mastery level — color-coded: red (below 40% average stability), yellow (40-70%), green (above 70%)
- Number of activities inside the module
- A "Review Now" shortcut button if this module has FSRS-scheduled cards due today

Clicking a module opens the **Module Detail view** — a sub-page within the Journey that shows all activities inside that module, their individual completion and mastery status, and a button to start or continue each one.

The student can add a new custom module directly from this view using a "+ Add Module" button at the bottom of the module list.

Activity Management

Within each Module Detail view, activities are listed with their type, last session date, and current mastery status. The student can:

- Start or resume any activity
- Edit activity content manually — add or modify individual flashcards, edit practice questions, adjust a Learning Guide's content
- Ask the AI to regenerate or expand an activity — "Generate 10 more practice questions on this module" or "Add more detail to the Learning Guide's section on Le Chatelier's Principle"
- Add a new activity type to a module — e.g., adding a Feynman session to a module that currently only has flashcards
- Delete an activity

Source Material

A collapsible section on the Journey Detail page shows all source material attached to the Journey — uploaded files, links, and text pastes. The student can add new source material at any time. When new material is added, Veridian prompts: "This new material covers [detected topics]. Would you like to add new modules or expand existing ones?" The student approves or dismisses the suggestion.

Journey Settings

A settings section at the bottom of the Journey Detail page (or accessible via the Edit button) allows the student to:

- Rename the Journey
 - Change the deadline
 - Change the privacy setting (Private or Public)
 - Archive the Journey
 - Delete the Journey — with a confirmation step that warns the student their FSRS history and activity data will be permanently lost
-

Sharing a Journey

A Journey can be shared in two ways, both accessible from the Share button on the Journey Detail page:

Private Link

Generates a shareable URL that opens the Journey in read-only preview mode for anyone with the link. The recipient can clone the Journey into their own account, after which it becomes their own independent copy with a fresh FSRS schedule and no connection to the original. Private links do not require the recipient to have a Veridian account to preview the Journey, but cloning requires sign-up.

Public — Community Library

Setting a Journey to Public publishes it to the Community Library where any Veridian user can find and clone it. A Journey must meet a minimum quality threshold to be published — it must have at least 3 modules and at least one complete activity per module. When published, the Journey displays the creator's username and a clone count. The creator can unpublish at any time, which removes it from the Community Library but does not affect copies already cloned by other students.

Privacy defaults to **Private** on creation. Students must actively switch to Public to publish. This prevents accidental publishing of incomplete or personal Journeys.

Editing an Existing Journey

Students can modify a Journey at any point after creation without losing their progress or FSRS history. Editable elements include:

- Journey name and subject label

- Exam deadline — changing this recalibrates the entire FSRS schedule forward or backward automatically
- Source material — add new files, links, or text at any time
- Modules — rename, add, delete, or reorder modules. Deleting a module with existing activity data prompts a warning
- Privacy setting
- Study mode preferences per module — e.g., switching a module from primarily flashcard-based to primarily practice quiz-based

What cannot be changed after creation without consequence: the diagnostic results. If a student wants to re-run the diagnostic for a module (because their knowledge has changed significantly), they can trigger a re-diagnostic from the Module Detail view, which will re-assess and potentially re-stage that module.

Community

Overview

The Community Library is where students discover, browse, and clone Journeys created and published by other Veridian users. It is the social distribution engine of the product — the mechanism by which one student's well-built Journey becomes useful to hundreds of others studying the same material. It is not a social feed, not a forum, and not a place to post or perform. It is a knowledge commons: a growing, community-built catalog of structured learning Journeys across every subject Veridian serves.

The Community Library is accessible to anyone — including visitors who do not yet have a Veridian account. Browsing and previewing Journeys requires no sign-in. Cloning a Journey requires an account. This is intentional: a student who receives a shared Journey link should be able to evaluate it before being asked to create an account, reducing friction at the point of acquisition.

Page Layout

The Community Library page has three primary elements arranged vertically:

1. **Search and filter bar** at the top
2. **Journey card grid** filling the main content area
3. **Individual Journey preview page** navigated to on click

There are no curated sections, featured picks, or editorial placements in the initial release. Discovery is driven entirely by search, filters, and sort order.

Search, Filters, and Sort

Search Bar

A prominent free-text search bar sits at the top of the page. Search matches against Journey titles, subject names, module names, and tags. Results update as the student types. An empty search state shows the full library sorted by the default sort order (most cloned).

Filters

A filter panel sits below the search bar, either as a horizontal row of filter chips on desktop or a collapsible filter drawer on mobile. Available filters:

- **Subject category** — broad groupings: AP Courses, Pre-Med, SAT/ACT, College Courses, Self-Learning, Language Learning, Other
- **Specific subject tag** — AP Chemistry, AP US History, Organic Chemistry, Biochemistry, etc. Tags are applied by the creator when publishing. Multiple tags can be selected simultaneously
- **Module count** — filter by Journey size: Small (1-3 modules), Medium (4-6 modules), Large (7+ modules)
- **Clone count range** — filter by popularity: any, 10+, 50+, 100+

Filters are additive — applying multiple filters narrows results. A clear all filters option resets to the full library.

Sort Dropdown

A sort dropdown sits at the top right of the Journey card grid. Sort options:

- **Most popular** (default) — sorted by total clone count, descending
 - **Highest rated** — sorted by net thumbs up ratio among users who have rated the Journey
 - **Most recent** — sorted by publish date, newest first
 - **Relevance** — only available when a search query is active. Ranks by how closely the Journey matches the search terms
-

Journey Cards in the Library

Each Journey in the library is represented by a card in the grid — two columns on desktop, one column on mobile. Each card displays:

- **Journey title and subject**
- **Tags** — up to 3 tags shown, additional tags truncated with a "+N more" indicator
- **Module count** — "6 modules"
- **Clone count** — "Cloned 214 times"
- **Rating summary** — a thumbs up percentage: "91% positive (48 ratings)"
- **Creator username** — "by @username"
- **Published date** — "Published 3 weeks ago"

Clicking anywhere on a card navigates to the full Journey Preview page for that Journey.

Journey Preview Page

The Journey Preview page is a full dedicated page with its own shareable URL. This means a student can send a direct link to a specific Journey and the recipient lands on the preview page immediately, whether or not they have a Veridian account.

Header

- Journey title and subject, large and prominent
- Tags displayed as colored pills
- Clone count and rating summary
- Creator username
- A prominent **Clone Journey** button — the primary action on the page
- A secondary **Share** button that copies the page URL to clipboard

What Is Publicly Visible

The preview page shows the structure and content of the Journey as the creator built it, with the following rules about what is and is not visible:

Always visible:

- All module names and their one-sentence descriptions
- The full activity list per module — what type each activity is (Learning Guide, Practice Quiz, Flashcard Set, Mastery Check) and how many items each contains
- Tags applied by the creator
- Clone count, rating summary, and published date
- Creator username

Never visible:

- The creator's personal exam deadline
- The creator's diagnostic results or module stage assignments
- The creator's personal progress, mastery scores, or FSRS history
- Any personal notes or annotations the creator may have added

Creator-controlled visibility:

- **Source material** — hidden by default. The creator can toggle source visibility on when publishing if they want to show what material the Journey was built from (e.g., "Built from Campbell Biology 12th Ed, Chapters 4-9"). Sources are never shown unless the creator explicitly enables this. This respects that some source

material is personal (handwritten notes, internal school documents) while allowing creators who used well-known references to signal credibility.

Module and Activity Preview

Below the header, all modules are listed in order. Each module is expandable to show its full activity list. Within the preview:

- Module names and descriptions are fully visible
- Activity types and item counts are visible — "Practice Quiz — 24 questions," "Flashcard Set — 67 cards"
- Actual question and card content is not visible in the preview — the student must clone the Journey to access the content itself

This gives enough information to evaluate the quality and scope of a Journey without giving away the full content, which protects the value of cloning and incentivizes the creator to build well-structured Journeys.

Rating and Reviews

Below the module list, a ratings section shows the aggregate thumbs up / thumbs down split and any user-submitted tags. Tags are short descriptors that raters can apply when rating: "Accurate," "Well structured," "Too basic," "Too advanced," "Great for AP exam," "Needs more practice questions." Multiple tags can be selected. The most commonly applied tags across all raters are displayed on the preview page as social proof signals.

Students who have cloned and used the Journey for at least 3 days are prompted once to leave a rating. The prompt appears inside the Journey Detail page after the 3-day threshold, not as a popup. Rating is always optional.

The Clone Flow

When a student clicks **Clone Journey** — either from the card grid or the preview page — a lightweight modal appears before the Journey is added to their account. The modal has two inputs:

1. **Exam or target date** — a date picker, required. The same field as in standard Journey creation. The cloned Journey's entire FSRS schedule will be calibrated to this date
2. **Module selection** — a checklist of all modules in the Journey with all boxes checked by default. The student can uncheck any modules they do not want

included. A note below the checklist reads: "You can always add or remove modules after cloning"

A **Clone** button at the bottom of the modal confirms the action. The Journey is immediately added to the student's account and they are taken to the Journey Detail page.

If the student is not logged in when they click Clone, they are prompted to sign in or create an account first. After authentication they are returned to the preview page and the clone modal opens automatically — they do not lose their place.

Diagnostic After Cloning

After cloning, the diagnostic runs the same as it does for any newly created Journey. The cloned Journey's modules and activities are pre-built, but the student's knowledge level for those modules is unknown to Veridian. The diagnostic assigns each module a starting stage based on the student's actual demonstrated knowledge — Stage A, B, or C — regardless of what stage the original creator was at when they published.

The diagnostic prompt on a cloned Journey is framed slightly differently to acknowledge the context:

"This Journey was built by @username. Before you dive in, let's see what you already know — 15 quick questions across the modules you selected. This takes about 5 minutes and means you won't start from scratch on things you already understand."

The skip option is present with the same benefit statement as in standard Journey creation.

Publishing a Journey

Publishing is managed from the Journey Detail page, not from the Community Library itself. A student sets their Journey to Public in the Journey settings, which triggers a brief publishing checklist:

- Journey must have at least 3 modules
- Each module must have at least one complete activity
- At least one tag must be applied

If the Journey does not meet these requirements, the student is shown which conditions are unmet with a direct link to fix each one. Once published, the Journey appears in the Community Library immediately.

The creator can unpublish at any time from the Journey settings. Unpublishing removes the Journey from the library for new browsers but does not affect copies already cloned by other students — those copies are fully independent and continue to exist in their owners' accounts.

Tags

Tags are applied by the creator during publishing and can be edited afterward. Tags serve as the primary discovery mechanism in the library alongside free-text search. A Journey can have up to 8 tags. Tags are a combination of:

- **Exam or course type** — AP Chemistry, AP US History, MCAT, Orgo 1, Biochemistry, SAT Math, etc.
- **Scope descriptor** — Full Course, Single Unit, Exam Prep, Quick Review
- **Level descriptor** — Beginner, Intermediate, Advanced

Veridian suggests tags automatically based on the Journey's subject and module content. The creator can accept, remove, or add to the suggestions before publishing.

Profile

Overview

The Profile & Settings page is the student's personal control center inside Veridian. It serves two distinct purposes: managing the student's account and surfacing the long-term learning intelligence Veridian has accumulated over time. These two functions live inside the same page because both are personal, persistent, and user-specific, but they are separated into clear tabs so the page never feels cluttered or confused.

This page is not a public profile, social identity surface, or gamification hub. It exists to help the student control how Veridian works for them and to understand what Veridian has learned about their studying over time. All data shown here is private to the student.

Page Layout

The Profile & Settings page is accessed from the left sidebar under **Profile**. It opens as a full page, not a modal. The page is structured around two top-level tabs:

- **Account**
- **Learning Profile**

The selected tab is shown directly under the page header. On desktop, the layout uses a centered content column with strong section spacing. On mobile, the same content stacks vertically with collapsible subsections where necessary.

The page header contains:

- The student's avatar
- Username / display name
- Joined date
- A short subtitle: "Your account and learning profile"

No public follower counts, no visible status badges, and no social metrics appear anywhere on this page.

Tab 1 — Account

Purpose

The Account tab contains everything related to identity, preferences, privacy, and account control. It is administrative and operational, not reflective. A student should be able to visit this tab and quickly change how Veridian behaves for them without having to think about their study data.

Section A — Identity

This section contains the student's editable identity fields:

- **Display name**
- **Username**
- **Avatar**

Display Name

A simple text field representing the name shown internally on the account. This may be the student's real first name, initials, or any preferred label.

Username

A unique handle used for authorship attribution in the Community Library when a student publishes a Journey. This is the only identity field visible to other users in public Journey contexts. It must be unique across the platform.

Avatar

A simple avatar selector. In the initial version, avatars are system-provided only. No custom photo uploads. This keeps moderation simple, avoids inappropriate imagery, and keeps the product clean.

Changes to any identity field save immediately or via an explicit save button depending on implementation, but the UI should make the save state obvious.

Section B — Account Access

This section displays the student's sign-in and authentication information.

It includes:

- **Primary email address**
- **Authentication provider** — Google, Apple, or email/password
- **Connected accounts** — if the student linked more than one sign-in method

The student can connect or disconnect authentication providers here, as long as at least one valid login method remains on the account at all times. This section is informational first and editable second — its job is to reassure the student about how they access their account and give them control if needed.

Section C — Notification Preferences

This section controls how Veridian reminds the student about due work and approaching deadlines.

Notification Options

The initial version supports email preferences only. The student can choose one of three options:

- **Daily Digest** — a daily summary of what is due that day
- **Urgent Only** — only alerts when work is overdue or a deadline is approaching
- **Off** — no email notifications

The Account tab is where this preference is managed. The Home screen may show a badge when work is due, but all notification control lives here.

A short explanatory note should appear beneath the selection:

"Veridian only sends reminders related to your Journeys and deadlines. You can change this anytime."

There is no need for highly granular notification settings in v1. Simplicity is better.

Section D — Privacy Defaults

This section controls how newly created and published Journeys behave by default.

Default Journey Privacy

A toggle or selection input determines whether new Journeys default to:

- **Private**

- **Public**

Private should be the default platform setting for all users, but the student can change their own default here if they routinely build public community Journeys.

Source Visibility Default

A second setting controls whether source material should be shown publicly by default when publishing a Journey.

Options:

- **Hide sources by default**
- **Show sources by default**

This setting only affects future published Journeys. It does not retroactively alter existing ones.

A note beneath this section should explain:

"Your personal deadlines, progress, diagnostic results, and learning history are never public. Only Journey structure and content are shared when you publish."

This is important for trust.

Section E — Account Controls

This section holds the higher-risk or irreversible actions.

Sign Out

A simple sign-out action.

Delete Account

A destructive action at the bottom of the page, visually separated from the rest of the tab.

Deleting the account should require:

1. Confirmation modal
2. Secondary warning about permanent loss of Journeys, learning history, and account settings

3. Final confirmation action

This action permanently removes the account and all associated private learning data. It should feel serious and deliberate, not easy to trigger by accident.

There is no need for account export in the first version.

Tab 2 — Learning Profile

Purpose

The Learning Profile tab is Veridian's long-term memory of the student's studying. It is the page where Veridian turns raw study activity into interpretable insight. This is not where the student changes settings; this is where they understand what subjects they are actually improving in, what they are neglecting, and how their knowledge is evolving across time.

This tab is global across all Journeys. Journey-specific insights remain inside each Journey's detail page. The Learning Profile exists to answer the bigger question: "What does my overall learning picture look like?"

Section A — Global Knowledge Map

This is the core visual of the Learning Profile tab.

The Global Knowledge Map is not a graph with floating nodes. It is a clean, structured list or grid of all modules the student has studied across every Journey. Each module is shown as a compact card or row with:

- Module name
- Parent Journey / subject
- Current mastery level
- Last reviewed date
- Current stage status if relevant

Mastery is color-coded:

- **Red** — weak / unstable
- **Yellow** — moderate / improving
- **Green** — strong / stable

This section allows the student to instantly scan where they stand across all ongoing and past study areas. It should be filterable by subject so that a student with many Journeys can narrow the view.

Clicking a module from this map should deep-link back into the relevant Journey or Module Detail page.

The purpose of this section is not visual novelty. It is clarity.

Section B — Subject Breakdown

This section gives a higher-level view of how the student's studying is distributed across subjects.

For each subject represented in the student's Journeys, Veridian shows:

- Total time studied
- Number of active modules
- Average mastery level
- Number of sessions completed
- Whether the subject is trending upward, stable, or at risk

Examples of subjects might include:

- AP Chemistry
- AP US History
- Organic Chemistry
- Biology
- MCAT Biochemistry

This section helps the student see whether their effort is balanced or lopsided. It also surfaces neglected subjects that may still have active deadlines.

The presentation should stay compact — most likely a vertical list of subject cards rather than a dense dashboard of charts.

Section C — Study History

This section is the chronological log of the student's actual work inside Veridian.

Each session entry includes:

- Date
- Journey name
- Module name
- Activity type — Learning Guide, Practice Quiz, Flashcards, Mastery Check
- Session duration
- Short outcome summary — for example:
 - "12 cards reviewed"
 - "18 question quiz completed"
 - "Mastery Check passed"
 - "Accuracy improved from 62% to 74%"

This is not just a timestamp log. It should help the student reconstruct what they have actually done. If they feel like they have not been working, this section provides a factual record. If they want to find their last session in a specific module, this gives them an easy path.

The default view should show the most recent sessions first, with a "Load more" pattern rather than full pagination.

Section D — Trends and Learning Signals

This section translates accumulated study behavior into readable patterns.

It should include a small number of high-value signals, not a wall of analytics.

Recommended signals:

- **Most improved subject** — where the student's average mastery has increased the most recently
- **Most neglected subject** — where work has not been done despite active modules or upcoming deadlines
- **Most stable subject** — where mastery remains consistently high over time
- **Most fragile area** — modules or subjects where retention is slipping and review is needed soon

These can be shown as summary cards or short insight rows.

This is where Veridian demonstrates intelligence. The student should feel that the system is not merely recording activity but actually understanding it.

Section E — FSRS Health Summary

This section gives the student a simplified view of how healthy their study system is overall.

It should not expose raw FSRS internals in a way that feels academic or confusing. Instead, it should translate them into useful states such as:

- **Strong** — most of your active material is stable
- **Needs attention** — some modules are becoming unstable
- **At risk** — overdue work is beginning to weaken retention

The section can also show a few high-level values such as:

- Active modules currently stable
- Modules at risk of slipping
- Overdue review load

The goal is not to teach the student the FSRS formula. The goal is to tell them whether their learning system is healthy and where pressure is building.

Data Boundaries and Privacy

Everything inside the Learning Profile tab is private. None of it is publicly visible through usernames, published Journeys, or shared links.

Specifically, the following are always private:

- Diagnostic results
- Module stage placements based on diagnostic performance
- Mastery levels
- Session history
- Subject trends
- FSRS health state
- Time studied
- Last reviewed dates tied to personal use

Publishing a Journey never publishes the creator's personal learning profile. It only publishes the reusable Journey structure and content.

This distinction should be made explicit inside the settings UI where relevant, because it is a core trust boundary in the product.

Design Notes

The Profile & Settings page should feel calm, private, and useful. It is the least performative page in the product. There should be no loud growth mechanics, no social proof widgets, and no unnecessary visual clutter.

The **Account** tab should feel controlled and minimal.

The **Learning Profile** tab should feel intelligent and reflective.

Most importantly, the page should not overwhelm the student with numbers for the sake of numbers. Every element should answer one of four questions:

- Who am I in Veridian?
- How is Veridian configured for me?
- What have I actually learned?
- Where do I need to focus next?

If a component does not help answer one of those, it likely does not belong on this page.

Study Mode

Overview

Study Modes are the actual learning activities inside Veridian. They are the actions a student takes within a Module once a Journey has been created, source material has been processed, and the diagnostic has assigned an appropriate starting point. If the Journey is the container and the Module is the unit of organization, the Study Modes are the engine of the product. This is where Veridian stops being a planning tool and becomes a learning system.

Every Module can contain multiple Study Modes. These are generated automatically from the source material and knowledge map when the Journey is created, but students can edit, expand, or regenerate them manually at any time. Veridian's job is to decide what kind of activity is most useful at a given moment. The student's job is to engage with that activity. This keeps the system frictionless without making it rigid.

The product is built around three stages of learning:

- **Stage A — Learn**
- **Stage B — Practice**
- **Stage C — Mastery**

These stages are not cosmetic labels. They represent meaningfully different cognitive tasks. Stage A is for first exposure and comprehension. Stage B is for retrieval, correction, reinforcement, and speed. Stage C is for proving understanding without support.

In addition to the stage-based structure, Veridian includes one specialized mode:

- **Cram Mode** — a high-intensity review mode available at any time, designed for deadline pressure

All Study Modes must support clean formatting across subjects. LaTeX rendering is required everywhere math or symbolic notation may appear, including Learning Guides, practice questions, flashcards, Feynman prompts, Free Recall prompts, and end-of-session summaries.

General Principles Across All Study Modes

Before outlining the modes individually, several rules apply across the entire study system.

1. Every Mode Ends With a Summary

Every Study Mode ends with a structured session summary. The summary is not just a receipt. It is a diagnostic output that tells the student what happened, what patterns Veridian noticed, and what the most useful next step is.

A session summary should include some combination of:

- What was covered
- Time spent
- Accuracy or completion level where relevant
- Concepts that appeared weak
- Concepts that appeared strong
- Any repeated mistake patterns Veridian identified
- A single recommended next action

This summary feeds directly back into the Journey Detail page and the student's Learning Profile over time.

2. Every Mode Should Feel Intentional

The student should always understand why they are in a given mode. If they are in Stage A, it should be obvious that Veridian believes they need structured guidance before practice. If they are in Stage B, it should be clear that they are now in active retrieval mode. If they are in Stage C, the framing should communicate that this is about proving mastery, not just doing more questions.

3. Activity Variety Matters

No mode should feel like static text pasted into a box. Every mode should have internal variation — question types, interaction styles, pacing, support tools, and dynamic feedback — especially in subjects where fatigue and disengagement are likely.

4. Soft Guidance Beats Hard Enforcement

Veridian is designed to help students study effectively, not trap them in rigid gates. In general, the app should suggest, steer, and intervene intelligently rather than punish or hard-lock students unless absolutely necessary.

Stage A — Learn

Purpose

Stage A is the structured learning phase. It is used when a student enters a Module with weak or nonexistent demonstrated knowledge, either because the diagnostic placed them there or because they manually chose to begin from first principles. Stage A exists to give the student a real conceptual foundation before they are pushed into repeated retrieval practice.

This is not passive reading. The entire point of Stage A is to make learning feel guided, interactive, and coherent without becoming overwhelming. If it feels like a giant PDF, it has failed.

Core Structure

A Stage A activity is called a **Learning Guide**. Each Learning Guide is broken into multiple concept sections derived from the Module's source material and AI-extracted knowledge map. These sections are shown in a visible sidebar or progress strip so the student always knows where they are and what remains.

Each concept section contains:

- A concise explanation of the concept in plain language
- Definitions of critical terms
- One or more worked examples for quantitative or process-heavy topics
- At least one embedded check-in question
- A short transition into the next concept

The student progresses through the Learning Guide section by section rather than facing a continuous wall of text. This creates a sense of movement and reduces cognitive overload.

Learning Guide Components

1. Concept Explanation

This is the explanatory core of the section. It should be clean, readable, and structured for comprehension rather than completeness. The purpose is not to reproduce a textbook chapter but to help the student actually understand what matters.

For every concept section, Veridian should prioritize:

- Plain-language explanation
- Correct terminology
- Clear distinction between similar concepts
- Concise structure over long prose

2. Worked Examples

For quantitative or procedural subjects — chemistry, math, physics, economics, statistics, organic chemistry — the Learning Guide must include worked examples.

Each worked example should show:

- The question or scenario
- The relevant formula or principle
- Step-by-step reasoning
- The final answer
- A brief explanation of why each step matters

Worked examples are not optional in these subjects. They are a core part of how students learn to move from theory to execution.

3. Embedded Check-In Questions

After each major concept section, the student gets a lightweight check-in. These are not full quizzes. They are short engagement prompts designed to confirm that the student followed the explanation.

Examples include:

- One multiple choice question
- One short-answer reflection
- A quick "explain the difference between X and Y" prompt
- A step-completion prompt inside a worked example

These questions should be easy enough to maintain flow but meaningful enough to detect whether the student is just scrolling.

4. Audio Narration

Every concept section can be played aloud with generated narration. This allows the Learning Guide to function more like an interactive lecture for students who prefer listening, multitasking, or auditory reinforcement.

The student can:

- Play the narration for a section
- Pause and resume
- Read while listening
- Move between sections while the guide remains in sync

Audio is not a gimmick. It makes Stage A feel less like reading a document and more like being taught.

5. External Review Suggestions

Each concept section should surface one or more suggested search prompts for external review content. These are not embedded videos or scraped links. They are AI-generated keyword suggestions the student can use to search YouTube or the web.

Examples:

- "Le Chatelier's Principle Khan Academy"
- "SN1 vs SN2 mechanism YouTube"
- "AP Bio cellular respiration overview"

This helps the student triangulate understanding without Veridian needing to host or control third-party content.

Stage A Interaction Model

The student moves through the Learning Guide section by section. Each section can be marked complete either automatically after reading and answering the embedded check-in or manually by the student.

If the student feels a section is too basic, they may skip that section. If they feel the whole Module was placed too low, they may skip directly to Stage B for that Module. Veridian should allow this, but gently note that Stage A was recommended based on their diagnostic.

The student should always feel guided, never trapped.

Stage A Summary

At the end of a Learning Guide session, Veridian generates a summary that includes:

- Sections completed
- Check-in performance
- Concepts that appear understood
- Concepts that may still be shaky
- Suggested next step — continue Stage A, begin Stage B Practice Quiz, or review a specific section again

The summary should also update the Module's visible strong/weak concept indicators inside the Journey Detail page.

Stage B — Practice

Purpose

Stage B is the active retrieval phase. This is the core of Veridian. The student is no longer just being introduced to the material — they are being asked to produce, discriminate, apply, and reinforce knowledge under varying conditions. Stage B is where real learning gains are most visible and where most students will spend the majority of their time.

Stage B contains four distinct activity types:

- Practice Quiz
- Flashcard Mode
- Interleaved Review
- Full Journey Challenge

Each activity serves a different purpose inside the same stage.

Stage B.1 — Practice Quiz

Purpose

The Practice Quiz is the default active practice activity for a Module. It is designed to generate fresh, AI-created questions based on the Module's source material and knowledge map. It adapts to the student's weaknesses, supports multiple question types, and provides meaningful post-session analysis.

This is not a one-size-fits-all quiz. It is configurable, adaptive, and deeply tied to the student's actual performance.

Quiz Setup

Before starting a Practice Quiz, the student chooses a setup. This setup should be fast, not overwhelming.

Question Count

The student chooses how many questions they want:

- 10 questions
- 25 questions
- 50 questions
- Custom amount

The system displays an estimated session time based on the student's typical pace.

Focus Preset

The student chooses one of three quiz focuses:

- **Full Review** — balanced coverage across the Module
- **Weak Spots** — emphasizes concepts with low mastery, repeated misses, or unstable retention
- **New Material** — emphasizes concepts the student has not practiced enough yet

If the student does not want to choose, Veridian can auto-select the recommended preset.

Timing Mode

The default mode is untimed. Each question has a rolling stopwatch showing how long the student took, but there is no overall time limit.

The student may optionally enable **Timed Mode**, which imposes an overall session clock. This is for exam simulation or speed practice. It is not the default because Stage B is primarily a learning environment, not a stress environment.

Question Types

Practice Quiz must support varied question types, including:

- Multiple choice
- Short answer
- True / false
- Multi-select
- Scenario-based questions
- Calculation-heavy questions with full LaTeX support
- Diagram-based questions if source material includes visual content

This variety prevents the student from memorizing interaction patterns instead of learning concepts.

Serving Logic

The Practice Quiz engine should prioritize questions using the following hierarchy:

1. Concepts or cards that are due for reinforcement
2. Concepts with low accuracy or repeated mistakes
3. Concepts that have not yet received enough practice
4. General balanced review when no urgent signal dominates

This keeps the quiz adaptive without making it unpredictable.

Mid-Quiz Soft Intervention

If the student gets multiple questions wrong in a row on the same concept, Veridian triggers a soft intervention.

This is not a hard stop or punishment. Instead, an inline prompt appears:

"You've missed a few questions on [concept]. Want a quick refresher before continuing?"

The student can choose:

- **Show me**
- **Keep going**

If they choose "Show me," Veridian surfaces a short concept reminder inline — usually a compact recap with one clarifying example. The student then returns to the quiz.

This is an important feature because it helps correct misconceptions while they are still active, without destroying flow or making the quiz feel punitive.

Practice Quiz Summary

At the end of the quiz, the student sees a structured report including:

- Questions completed
- Accuracy
- Average time per question
- Strongest concepts
- Weakest concepts
- Repeated error patterns
- Whether the student struggled more with speed, recognition, or concept transfer
- Recommended next action

Examples of recommendations:

- Review the Learning Guide section on equilibrium before doing another quiz
- Run a Weak Spots session next
- Move to Flashcard Mode for terminology-heavy content
- You're ready for a Mastery Check

This summary also updates the Module's concept-level strength and weakness indicators.

Stage B.2 — Flashcard Mode

Purpose

Flashcard Mode is the memory reinforcement layer of Veridian. It is driven by FSRS and is responsible for making sure facts, definitions, formulas, and recognition-based material are revisited at the right moment before they are forgotten.

This mode is especially useful for:

- Vocabulary-heavy content
 - Equations and formulas
 - Definitions
 - Image identification
 - Term-concept associations
-

Card Structure

Cards are generated automatically from source material during Journey creation, but the student can:

- Edit any card manually
- Add cards manually
- Ask the AI to generate more cards
- Delete cards that feel low value
- Add images or formulas where relevant

Cards can be front/back, image-based, or concept-prompt style depending on the material.

Review Flow

A card is shown. The student attempts recall. Then they reveal the answer and self-rate how well they knew it using standard FSRS-style recall quality buttons:

- Again
- Hard
- Good
- Easy

Veridian uses this rating to determine the next review interval.

The student should not need to understand the scheduling math. They just need to rate honestly.

Flashcard Summary

The summary includes:

- Cards reviewed
- Cards marked Again / Hard / Good / Easy
- Which concepts are stabilizing
- Which concepts remain fragile
- Recommended next step

This mode is a quiet but critical part of the whole system because it powers long-term retention and drives Due Today recommendations on the Home page.

Stage B.3 — Interleaved Review

Purpose

Interleaved Review is a mixed-practice mode that pulls questions from multiple Modules within the same Journey instead of focusing on one Module at a time. This is intentionally harder. It creates desirable difficulty and improves long-term retention by forcing the student to discriminate between concepts rather than staying inside one predictable lane.

This mode exists because real exams rarely isolate concepts as cleanly as practice sessions do.

Setup

The student chooses:

- Number of questions
- Which Modules to include
- Timed or untimed mode

By default, Veridian recommends Modules that are both important and unstable.

Use Case

Interleaved Review is most useful when:

- A student is preparing for a cumulative exam
 - Multiple Modules are already in Stage B
 - The student wants to test discrimination and transfer, not just repetition
-

Interleaved Summary

The summary emphasizes:

- Which Modules held up well when mixed together
- Which Modules collapsed under mixed conditions
- Whether the student is confusing similar concepts across Modules

This is one of the most scientifically valuable modes in the product and should be explained clearly in the UI so students understand why it feels harder.

Stage B.4 — Full Journey Challenge

Purpose

The Full Journey Challenge is the closest thing Veridian has to a course-wide or exam-wide benchmark. It pulls from every Module in the current Journey and gives the student a broad snapshot of where their overall preparation stands.

This is not just another Practice Quiz. It is a larger, more representative check on full-scope readiness.

This mode is especially useful for:

- AP exam preparation
 - Midterm or final prep
 - Seeing whether one or two strong Modules are masking broader weakness
-

Setup

The student chooses:

- Challenge length — short, medium, long

- Timed or untimed
- Optional weighting — balanced across all Modules or slightly biased toward weak areas

The default should be balanced across all Modules so the result reflects overall readiness, not just current weaknesses.

Output

The Full Journey Challenge summary should include:

- Overall accuracy
- Module-by-module breakdown
- Weakest Modules
- Strongest Modules
- Overall readiness estimate
- A recommended study plan for what to tackle next inside the Journey

This is one of the clearest measures of "how is my test prep actually going?" and should be framed that way.

Stage C — Mastery

Purpose

Stage C is where the student proves genuine understanding. This stage is not about repeated recognition or routine recall. It is about reconstruction, articulation, and transfer. The student should be able to explain, retrieve, and connect concepts without being handheld by familiar prompts.

Stage C contains three activities:

- Feynman Mode
- Free Recall
- Synthesis Questions

All three are worth keeping. They test different dimensions of mastery and complement one another.

Stage C.1 — Feynman Mode

Purpose

Feynman Mode asks the student to explain a concept in their own words. This is one of the best ways to detect shallow understanding. A student who "knows" a concept often discovers very quickly, when forced to explain it clearly, where their understanding breaks down.

Input Options

The student can complete Feynman Mode by:

- Typing their explanation
- Speaking their explanation with voice input

Voice support is especially important here because explanation feels more natural aloud than in essay form.

AI Response

Veridian analyzes the explanation and identifies:

- Missing pieces
- Vagueness
- Misconceptions
- The weakest specific point in the explanation

Rather than just assigning a score, it should surface the weakest point and ask a targeted follow-up question. This makes the mode feel like intelligent tutoring rather than grading.

Feynman Summary

The summary should include:

- Concept explained
- Confidence of explanation

- Weakest point identified
 - Suggested follow-up concept to review
 - Whether the student appears ready for synthesis-level practice
-

Stage C.2 — Free Recall

Purpose

Free Recall asks the student to reconstruct what they know from memory without prompts. This is one of the purest tests of whether learning actually stuck.

The student is shown a topic, module, or concept prompt and asked to write or speak everything they can recall about it.

Hint System

A blank page can create paralysis, so Free Recall must include a **Hint** button.

Hints should be tiered:

- **Hint 1** — reveals a section name or subtopic
- **Hint 2** — reveals a key term
- **Hint 3** — reveals a one-sentence conceptual nudge

Hints do not end the session or invalidate it. They are support tools. The number and depth of hints used should be reflected in the summary so the student sees where independence was strong or weak.

AI Grading

Veridian compares the student's recall against the Module's knowledge map and identifies:

- Concepts covered correctly
 - Important concepts omitted
 - Inaccuracies or misunderstandings
 - Coverage completeness
-

Free Recall Summary

The summary includes:

- Coverage estimate
- Missed ideas
- Incorrect ideas
- Hints used
- Next concept to revisit

This mode is extremely valuable because it reveals not just whether the student can recognize information, but whether they can produce it unprompted.

Stage C.3 — Synthesis Questions

Purpose

Synthesis Questions are the highest-difficulty activity in Veridian. They combine concepts across multiple Modules within the same Journey and ask the student to apply them together. This is where isolated knowledge becomes integrated understanding.

For v1, synthesis is limited to the current Journey only. It does not combine content across unrelated Journeys.

Use Case

This mode is best used when:

- The student is approaching an exam
 - Multiple Modules are already relatively stable
 - Veridian needs to test whether the student can connect ideas under realistic conditions
-

Question Design

Synthesis questions should require:

- Comparing concepts across Modules

- Applying one concept through the lens of another
- Choosing between multiple plausible frameworks
- Explaining why a certain principle applies in a mixed scenario

This mode should feel difficult in a productive way. If a student can do well here, they are close to real mastery.

Synthesis Summary

The summary should emphasize:

- Which Module combinations the student handled well
 - Where integration broke down
 - Which Modules are individually strong but weak when combined
-

Cram Mode

Purpose

Cram Mode is a high-intensity review mode built for urgency. It is always available manually, but especially useful when the student is close to a deadline and needs Veridian to compress the highest-value material into a focused session.

This is not a replacement for spaced repetition. It is an emergency mode for when ideal studying did not happen and the student still needs the best possible short-term outcome.

Cram Logic

Cram Mode prioritizes:

- Weak concepts
- Overdue reviews
- High-importance material
- Recently missed concepts
- Material most likely to improve short-term exam performance

It excludes Learning Guides by default. Cram Mode is about retrieval and correction, not deep first-time instruction.

Cram Experience

Cram sessions should feel focused, direct, and slightly more intense than normal Stage B practice. The student should not have to configure much. A single prompt like "Build my cram session" is enough, with Veridian doing the rest.

Optional toggles can include:

- Include flashcards
 - Include practice questions
 - Include interleaving
 - Timed or untimed
-

Cram Summary

The summary should be brutally clear and useful:

- What was covered
- What still looks dangerous
- What to do next if any time remains
- Which concepts must be reviewed again before the exam

This summary should feel like triage.

Final Notes

The Study Modes section is where Veridian becomes a real learning system rather than just a smart organizer. The value is not in any one activity alone, but in how the activities work together:

- Learning Guides build the foundation
- Practice Quizzes and Flashcards reinforce it
- Interleaving and Journey Challenges test broader readiness
- Feynman, Free Recall, and Synthesis prove genuine mastery
- Cram Mode handles real-world urgency when ideal plans fail

Together, these modes create a complete learning experience that is active, adaptive, and personalized — not just organized.

Tech Stack

Overview

Veridian is built as a client-heavy Single Page Application hosted on Base44. The architecture prioritizes build speed, developer simplicity, and zero backend management overhead — appropriate for a solo developer building a complex product without capital or a team. Every technology choice in this stack is either already present in the Axiom repository or is a deliberate addition to serve a specific product requirement.

This section documents what is used, why it is used, and where the important implementation decisions live.

Platform — Base44

Base44 is the foundational platform for the entire application. It replaces what would otherwise be a separate backend, database, authentication system, file storage layer, and hosting provider.

What Base44 handles:

- User authentication — sign up, sign in, session management, Google OAuth
- Database — entity storage and querying via the [@base44/sdk](#)
- File storage — PDF and document uploads from students
- Hosting and deployment — production environment and CDN
- Built-in LLM access — [InvokeLLm](#) is available but reserved for lightweight, low-stakes tasks only due to account limits

What Base44 does not handle:

- Heavy AI workloads — these route through the Gemini API directly
- FSRS scheduling logic — this runs client-side in JavaScript
- Background job processing — generation tasks happen synchronously with loading states in v1
- Full-text search at scale — Base44 querying is sufficient for v1 Community Library

The [base44/entities/](#) directory contains all data model definitions as [.jsonc](#) schema files. Every entity Veridian needs — Journeys, Modules, Activities, Cards, Sessions, Community Library entries — is defined here and auto-generates the corresponding SDK methods for reading and writing.

Frontend Framework

React 18 + Vite 6

React is the required framework for Base44 compatibility. Vite provides the build tooling — fast hot module replacement in development and optimized production builds.

The codebase uses `.jsx` files throughout. TypeScript is available in `devDependencies` and the `jsconfig.json` provides path aliases and type hinting, but the project is not strictly typed. This is acceptable for v1 speed.

Folder structure follows the Axiom convention:

```
text
src/
  api/          - Base44 SDK integration layer and Gemini API calls
  axiom/        - Core app logic and utilities
  components/   - Reusable UI components
  views/        - Page-level components tied to routes
  hooks/        - Custom React hooks
  store/        - Zustand global state stores
  utils/        - Pure utility functions including FSRS logic
  lib/          - shadcn utility setup and shared configuration
  css/          - Global styles
```

Routing

React Router DOM v6

All navigation inside Veridian uses React Router. Route structure maps to the product sections:

```
text
/          - Landing / waitlist (unauthenticated)
/home      - App Home (authenticated)
/journeys  - Journey list
/journeys/:id - Journey Detail
/journeys/:id/module/:moduleId - Module Detail
/study/:sessionId - Active study session (any mode)
/library   - Community Library
/library/:journeyId - Journey Preview page (public, no auth required)
```

/profile

– Profile & Settings

The `/library/:journeyId` route must be accessible without authentication for shareable public Journey links to work. All other routes redirect to sign-in if unauthenticated.

Styling and UI

Tailwind CSS v3 + Radix UI + shadcn component pattern

Tailwind handles all styling. No CSS-in-JS, no styled components. The full Radix UI primitive suite is installed — accordions, dialogs, dropdowns, tabs, tooltips, sliders, and more. These provide accessible, unstyled interactive logic that is then styled with Tailwind.

The `components.json` and shadcn pattern means UI components live in `src/components/ui/` and are fully owned and editable. This is the right approach — no black-box component library making visual decisions for you.

Supporting styling utilities:

- `class-variance-authority` — for component variant definitions
- `tailwind-merge` — for safely merging Tailwind classes without conflicts
- `tailwindcss-animate` — for simple CSS animations
- `next-themes` — for dark/light mode support

Veridian should be designed dark mode first. The aesthetic should be clean, modern, and minimal — not aggressive or cyberpunk. It should feel like a premium study tool, not a game.

Animation

Framer Motion v11

All meaningful UI transitions and animations use Framer Motion. This includes:

- Page transitions between routes
- Modal and sheet entrance and exit
- Study mode card transitions
- Journey creation step flow
- Session summary reveal animations

Framer Motion should be used with discipline — animation should reinforce the product, not perform. Every animation should have a clear purpose: confirming an action, providing orientation, or reducing the jarring quality of state changes.

Use the `motion` components from `framer-motion` and keep animations GPU-bound by relying on `transform` and `opacity` properties to prevent layout thrashing.

State Management

Two-layer state model:

Client State — Zustand

Zustand handles transient UI state that does not need to be persisted or fetched from the server. This includes:

- Current study session state — active question, timer, answer state, session progress
- Sidebar open/closed state
- Active modal and sheet state
- Quiz setup configuration before a session starts
- Diagnostic flow progress

Zustand stores live in `src/store/`. Each major feature area — study sessions, journey creation, UI state — should have its own store file rather than one monolithic store.

Server State — TanStack Query v5

TanStack Query manages all data fetched from Base44 — Journeys, Modules, Activities, Cards, Library entries, session history. It provides:

- Automatic background refetching
- Stale-while-revalidate caching so navigating back to a page feels instant
- Optimistic updates for actions like completing a card review or marking a section complete
- Request deduplication

Every Base44 entity fetch should go through a TanStack Query hook. Raw SDK calls should not be scattered across components.

Forms and Validation

React Hook Form + Zod

All forms — Journey creation, account settings, quiz setup, publishing flow — use React Hook Form for state and Zod for schema validation. This combination handles complex multi-step forms efficiently without re-rendering the entire form on every keystroke.

Zod schemas should be defined in `src/utils/schemas/` and shared between form validation and any runtime data validation needed from AI responses.

AI Integration — Gemini API

The Gemini API is the intelligence layer of Veridian. All AI-powered features route through Gemini. Base44's built-in `InvokeLLm` is available but used only for lightweight, non-critical tasks to stay within account limits.

Model selection:

- **Gemini 2.0 Flash** — default model for all generation tasks. Fast, cost-effective, strong instruction following, sufficient for structured JSON output
- **Gemini 2.0 Pro** — reserved for complex reasoning tasks: Feynman Mode explanation grading, Free Recall evaluation against the knowledge map, synthesis question generation. Use sparingly to control costs

All AI calls live in `src/api/ai/` — never scattered across components or hooks. Each AI function should be a clean, exported async function that accepts structured inputs and returns validated outputs.

Primary AI tasks and their prompt categories:

1. Document Ingestion and Knowledge Map Extraction

The student uploads a PDF or pastes text. The AI reads the document and returns a structured knowledge map — a JSON object listing all concepts, their relationships, key terms, and importance weighting.

Gemini 2.0 Flash accepts PDFs natively via the Files API. Upload the file, pass it with the extraction prompt, receive structured JSON. No external PDF parsing service needed.

The knowledge map is the foundation of everything else. The quality of every downstream activity depends on the quality of this extraction. This prompt must be the most carefully engineered in the entire system.

2. Learning Guide Generation

Takes the knowledge map and source content and produces a structured Learning Guide per module: concept sections, worked examples, embedded check-in questions, and external search keyword suggestions.

Output must be structured JSON that maps directly to the Learning Guide component's data model. The AI should not return free-form prose that needs parsing.

3. Practice Question Generation

Takes a module's knowledge map, a difficulty level, a question type specification, and existing question history to avoid repeats. Returns an array of question objects with correct answers, distractors, and explanations.

Questions must include LaTeX notation where mathematical or symbolic content is present. The prompt must explicitly instruct the model to use LaTeX delimiters consistently.

4. Flashcard Generation

Takes the knowledge map and returns front/back card pairs optimized for recognition-based recall. Cards should target atomic facts, definitions, and term-concept associations rather than broad explanations.

5. Feynman Mode Grading

Takes the student's typed or transcribed explanation and the module's knowledge map. Returns: an overall assessment, the specific weakest point in the explanation, and one targeted follow-up question focused on that weak point.

6. Free Recall Grading

Takes the student's brain-dump and the module's knowledge map. Returns: a coverage percentage, a list of concepts covered correctly, concepts missed, and concepts stated incorrectly.

7. Synthesis Question Generation

Takes multiple module knowledge maps from within the same Journey. Returns questions that require connecting concepts across those modules. These should be the highest difficulty questions in the system and should be validated for coherence before serving.

8. Post-Session Pattern Analysis

Takes session performance data — questions answered, accuracy per concept, time per question, hint usage. Returns a plain-language summary of patterns detected and a specific recommended next action.

General AI engineering rules:

Every Gemini call must:

- Request JSON output explicitly using Gemini's `responseMimeType: "application/json"` parameter
 - Include a Zod schema validation step on the response before it is used — AI output is never trusted raw
 - Have a fallback behavior if the response fails validation — retry once, then surface a graceful error to the user
 - Be wrapped in try/catch with meaningful error states shown in the UI
-

FSRS Implementation

FSRS (Free Spaced Repetition Scheduler) is the algorithm that determines when each flashcard is due for review. It runs entirely client-side in JavaScript.

Library: `ts-fsrs` — the reference TypeScript implementation of FSRS v5, usable in a JavaScript project. Install via npm.

Where it lives: `src/utils/fsrs/` — a clean wrapper around the `ts-fsrs` library that exposes the specific functions Veridian needs:

- `scheduleCard(card, rating)` — takes a card's current state and the student's recall rating, returns the next review date and updated stability/difficulty values
- `getDueCards(cards)` — filters a card set to return only cards due today or overdue
- `getCardHealth(card)` — returns a simplified health status for a card used in the Learning Profile FSRS summary

Card state — stability, difficulty, due date, review history — is stored on the card entity in Base44 and updated after every review session.

The Home screen's Due Today count is derived from `getDueCards` run against all active cards across all Journeys at session load.

LaTeX Rendering

LaTeX rendering is a non-negotiable requirement for math and science subjects. It must work consistently across every text surface in the application.

Library: `KaTeX` — faster and more reliable than MathJax for React applications. Install `katex` and `react-katex`.

Surfaces that must render LaTeX:

- Learning Guide concept sections and worked examples
- All practice question stems, options, and explanations
- Flashcard fronts and backs
- Feynman Mode input and AI feedback
- Free Recall prompt and grading output
- Session summaries
- Any user-editable text field that accepts module content

Create a shared `<LatexRenderer>` component that wraps KaTeX and is used everywhere text content is displayed. It should detect LaTeX delimiters automatically — both `\( \)` for inline and `\[ \]` for block — and render accordingly. All other text passes through unchanged.

The Gemini prompt for question generation must explicitly instruct the model to use these exact delimiters for all mathematical notation.

Voice Input — Feynman and Free Recall

Voice input is needed for Feynman Mode and optionally for Free Recall.

API: Web Speech API — browser-native, no external service or API key required. Available in all modern browsers including Chrome, Edge, and Safari.

Implementation:

- A microphone button activates `SpeechRecognition`
- Transcription streams into the text input in real time
- The student can correct the transcript before submitting

- If the browser does not support the Web Speech API, the voice button is hidden and the text input is the only option

This requires no backend, no cost, and no third-party dependency. The Web Speech API is the right tool here.

Audio Narration — Learning Guide

Learning Guide audio narration uses the **Web Speech Synthesis API** — also browser-native and zero cost. This handles text-to-speech for concept section narration.

For v1 this is sufficient. The voice quality is acceptable in modern browsers, especially on Chrome. A future upgrade would be Google Cloud Text-to-Speech for higher quality narration, but this is not necessary to launch.

Implementation:

- Each concept section has a play button
 - `SpeechSynthesis.speak()` reads the section text aloud
 - The student can pause, resume, or stop narration
 - Narration syncs to section — stopping and starting a new section cancels the previous
-

Rich Text and Markdown

react-markdown handles rendering of AI-generated content that comes back as Markdown. Learning Guide sections, explanations, and post-session summaries are all Markdown rendered through this component.

react-quill is available for any rich text editing surfaces where students write free-form content — Feynman Mode typed explanations, manual notes on modules. For v1 these surfaces can use plain textarea inputs. React Quill becomes relevant when richer formatting is needed.

Drag and Drop

@hello-pangea/dnd handles drag and drop reordering. Used in:

- Module reordering within Journey Detail
- Activity reordering within a Module

- Card reordering within Flashcard editor

This is a maintained fork of the original [react-beautiful-dnd](#) with React 18 compatibility.

Charts and Data Visualization

Recharts handles all data visualization in the Learning Profile tab. Specific uses:

- Subject breakdown — bar or horizontal bar chart
- Mastery trends over time — line chart
- FSRS health distribution — simple bar or area chart
- Session accuracy history — line chart

Recharts is already in the dependency tree. Keep charts minimal and purposeful — do not add charts for visual impressiveness. Every chart must answer one of the four questions defined in the Learning Profile section.

Date and Time

date-fns is the preferred date utility. It is tree-shakeable and more modern than moment.js. Both are installed — use date-fns for all new work and do not add new moment.js usage.

Date-fns is used for:

- Calculating days until deadline for Journey scheduling
 - Formatting session dates in Study History
 - Calculating FSRS due dates
 - Displaying relative time in the Community Library
-

Payments — Stripe

Stripe is already wired in via [@stripe/react-stripe-js](#) and [@stripe/stripe-js](#). While Veridian is free and has no paid tiers in v1, the infrastructure is present for future use — optional donations, institutional licensing, or grant-funded premium features if the product direction ever requires it. No active payment flows should be built in v1.

Notifications — Sonner

Both Sonner and react-hot-toast are installed. **Use Sonner exclusively.** It is more modern, has better default styling, and supports rich toast content. Remove react-hot-toast from active use to keep the notification layer consistent.

Sonner handles:

- Success confirmations — Journey created, session complete, Journey published
 - Error states — AI generation failed, upload failed
 - Informational toasts — auto-save confirmation, sync status
-

Icons

Lucide React is the icon library. Consistent throughout. Do not mix in other icon libraries.

Key Technical Rules for the Codebase

These are implementation standards that should be followed consistently:

1. **All AI calls go through `src/api/ai/`** — never call Gemini directly from a component or hook
 2. **All Base44 entity operations go through TanStack Query hooks** — never call the SDK directly from a component
 3. **All FSRs logic goes through `src/utils/fsrs/`** — never call ts-fsrs directly from a component
 4. **LaTeX rendering uses the shared `<LatexRenderer>` component** — never render LaTeX ad-hoc
 5. **Animations use Framer Motion** — no mix of CSS transitions and Framer Motion on the same element
 6. **Forms use React Hook Form + Zod** — no uncontrolled inputs or ad-hoc validation
 7. **Dates use date-fns** — no new moment.js usage
 8. **Notifications use Sonner** — not react-hot-toast
 9. **AI responses are validated with Zod before use** — never trust raw AI output
 10. **Public routes are explicitly defined in the router** — all unlisted routes redirect to auth
-

Third-Party Services Summary

Service	Purpose	Cost Model
Base44	Platform, auth, DB, hosting	Subscription tier
Gemini API (Flash)	Primary AI generation	Per token
Gemini API (Pro)	Complex reasoning tasks	Per token, use sparingly
Web Speech API	Voice input	Free, browser-native
Web Speech Synthesis	Audio narration	Free, browser-native
Stripe	Payment infrastructure, dormant in v1	Per transaction when active

Onboarding

Overview

Onboarding is the most important flow in the product. It is the difference between a student who understands what Veridian is and returns every day, and a student who signs up, sees something confusing, and never comes back. The goal of onboarding is not to show every feature. It is to get the student to one specific moment — the moment Veridian feels genuinely useful — as fast as possible.

That moment is completing their first study session inside a real Journey built from their actual material.

Everything in this document is designed to reach that moment without friction, without confusion, and without making the student feel like they are filling out paperwork.

Phase 1 — Unauthenticated Landing State

When a student arrives at Veridian for the first time without an account — from a shared Journey link, a word-of-mouth recommendation, or organic discovery — they land on the public-facing index route /.

This is not the full marketing landing page. That comes later when there is something real to show. For now the unauthenticated state is a minimal, focused page that answers three questions immediately:

- What is this
- Why should I care
- How do I start

Layout

The page should be clean and uncluttered. It contains:

- **Veridian wordmark and logo** — top left
- **Headline** — one line, large, direct. Something like: "Study smarter. Actually remember it." Not clever, not vague. Instantly understood
- **Sub-headline** — one to two sentences maximum. "Veridian builds personalized study journeys from your own material, then uses spaced repetition to make sure nothing gets forgotten before exam day."
- **Two calls to action:**
 - **Get Started** — primary button, takes the student to sign-up

- **Browse the Community Library** — secondary link, takes them to /library without requiring an account
- **One visual** — a clean screenshot or mockup of what a Journey looks like inside the app. Not an abstract illustration. Something real that shows the product.

The Community Library CTA matters. A skeptical student who is not yet ready to commit to an account should be able to see real published Journeys and evaluate the quality of the product before giving an email address. Lowering this barrier increases sign-up conversion.

Shared Journey Link State

If the student arrived via a shared Community Library Journey URL — /library/:journeyId — they land directly on that Journey's preview page, not the index. The preview page should have a persistent banner at the top:

"You're viewing a Community Journey. Create a free account to clone it and start studying."

The banner contains a single **Sign Up Free** button. It is not a modal gate. The student can read the full preview and explore the Journey structure without signing up.

Phase 2 — Sign Up Flow

The sign-up page lives at /signup. It should be the shortest possible version of account creation. Every additional field is friction that loses users.

Sign-Up Form

Step 1 — Authentication:

Two options presented with equal visual weight:

- **Continue with Google** — the primary option. One click, no typing, no friction. Should be visually prominent
- **Sign up with email** — secondary option. Opens an email and password input

For email sign-up:

- Email address
- Password — minimum 8 characters, one visible strength indicator, no complex requirements

- No confirm password field — this adds friction with minimal security value

That is all. No phone number, no school name, no grade level, no subject preferences at this stage.

Step 2 — Display Name (email sign-up only):

After entering email and password, one additional field:

- "What should we call you?" — a single first name or preferred name input

Google sign-up skips this step entirely because the display name is pulled from the Google account.

After Account Creation

The student is taken directly into the app. No email verification wall. If Base44 requires email verification, it appears as a dismissible soft banner inside the app:

"Verify your email to secure your account — check your inbox for a link."

This banner sits at the top of the screen and does not prevent access to any feature. It disappears permanently once the email is verified or manually dismissed.

Returning User Sign-In

Returning users sign in at /signin. Same layout as sign-up but with fewer fields. A returning user who has already completed onboarding lands directly on /home. A returning user who created an account but never completed onboarding — never built a Journey — is taken back into the onboarding welcome flow from the point they left off.

Phase 3 — Welcome Flow: Stepped Personalization Sequence

Immediately after sign-up, before seeing the app proper, the student enters a structured welcome sequence. This is a full-screen stepped flow — one question or prompt per screen — that feels conversational and fast, not bureaucratic.

The sequence has four steps. Progress is shown as a small dot indicator at the bottom of each screen. Each step has a visible **Skip for now** text link in the bottom corner. Nothing in this flow is mandatory.

Welcome Screen — Step 0 (no question, just orientation)

Before any questions, a brief warm welcome screen:

- Veridian logo centered
- Headline: "Welcome to Veridian, [first name]."
- Sub-headline: "Let's get you set up in about a minute. We'll personalize your experience and build your first study Journey together."
- Single button: **Let's go**

This screen exists to set expectations. The student knows what is about to happen and approximately how long it takes.

Step 1 — What Are You Studying For?

Prompt: "What are you studying for right now?"

Input: Large selectable chips — multi-select allowed. Options:

- AP Exams
- SAT / ACT
- Pre-Med
- College Courses
- Medical / Professional School (MCAT, LSAT, etc.)
- Self-Learning
- Something else

When the student selects chips they should visually activate — a clear selected state with color and a checkmark. Multiple can be selected simultaneously.

A text input appears below when "Something else" is selected.

Skip for now is available at the bottom. If skipped, the subject is set to null and the home screen shows generic suggestions.

Button: **Continue** appears once at least one chip is selected. If skipped, **Skip for now** serves as the navigation action.

Step 2 — When Is Your Next Exam?

Prompt: "When is your next big exam or deadline?"

Input: A date picker — calendar style, clean and simple. Below the date picker, a text alternative for students without a specific date: **"I don't have one yet"** — a tappable option that replaces the date picker selection.

This date is used to calibrate how urgently the home screen frames the student's workload and to pre-populate the deadline field in their first Journey.

Button: Continue

Step 3 — Where Are You Right Now?

Prompt: "How would you describe where you are with this material?"

Input: Three large selectable cards — single select:

- **Just starting out** — "I haven't really studied this yet"
- **Somewhat familiar** — "I know some of it but have gaps"
- **Final review mode** — "I know most of it and need to lock it in"

This answer is used to frame the diagnostic flow and set the tone of the home screen. A student who says "Just starting out" should not get the same framing as one who says "Final review mode."

Button: Continue

Step 4 — First Journey Prompt

This is the most important step in the entire onboarding sequence. The student has answered three quick questions. Now Veridian pivots from gathering information to doing something with it.

Prompt: Contextual based on Step 1 answer. Examples:

- If AP Exams selected: "Let's build your first AP study Journey. Do you have any material — class notes, a textbook chapter, a syllabus, or a past exam?"
- If Pre-Med selected: "Let's build your first pre-med Journey. Upload your lecture notes, a textbook chapter, or any study material you want to work from."
- If no subject selected / skipped: "Let's build your first Journey. Upload any study material — notes, a PDF, a textbook chapter — and we'll turn it into a personalized study plan."

Input options — two large options presented side by side or stacked:

- **Upload a file** — PDF, image, or document. Tapping this opens the file picker
- **Paste text** — a text area appears for the student to paste notes, a syllabus, or any raw text directly
- **Find one in the Community Library** — a third, slightly smaller option below the two primary ones. This is for students who do not have material ready. It takes them to the Community Library in a guided state where they can browse and clone a Journey instead of building one from source material

The **Find one in the Community Library** option solves a real problem: some students arrive at Veridian without having their study material immediately accessible. Forcing them to upload something before they can see the app in action is a barrier. Letting them clone a Community Journey instead means they still exit onboarding with something real set up, just via a different path.

If the student uploads material or pastes text, the standard Journey creation flow begins from this point — AI processing, knowledge map generation, module review, deadline setting, and diagnostic. This is the same flow described in Section 2 but entered from the onboarding context. The subject from Step 1 and the deadline from Step 2 are pre-populated.

If the student skips this step entirely via **Skip for now**, they are taken to the home screen in its empty state.

Phase 4 — First App Experience After Onboarding

After completing the welcome flow — whether they created a Journey, cloned one, or skipped — the student lands on the home screen at /home.

If they created or cloned a Journey, the home screen looks and functions exactly like the normal home screen described in Section 1. No special first-day framing, no tutorial overlay, no checklist widget. The product just works. The Journey is there. If cards are already due from the diagnostic setup, Due Today shows them. The student can start immediately.

If they skipped Journey creation, the home screen shows the empty state version with a strong persistent prompt to create their first Journey. This prompt should feel like a natural next step, not an error state:

"You're all set up. Now let's build your first Journey — upload your study material and Veridian will handle the rest."

A single **Create Journey** button sits below this message.

Phase 5 — Contextual Hint System

After onboarding the product teaches itself through contextual hints — small, dismissible tooltips or inline banners that appear the first time a student encounters a feature. There is no product tour, no feature checklist, and no walkthroughs. These patterns interrupt real use and signal that the product is hard to understand.

Instead, hints appear naturally in context, exactly when and where they are relevant.

Hint Rules

- Each hint appears **once per user, ever**. After it is dismissed or the student navigates away, it never appears again
- Hints are stored in Base44 on the user entity as an array of seen hint IDs — `onboarding.hintsShown: ["due_today", "fsrs_rating", "session_summary", ...]`
- Hints do not block the UI. They appear as non-modal callout bubbles, inline banners, or tooltip overlays. The student can always interact with the page beneath them
- Hints are dismissible with a single tap or click on an X button
- Hint copy should be one to two sentences maximum — explain what something is and why it matters, nothing more

Hint Catalogue

Home Screen — Due Today (first visit with a Journey set up):

"These are the cards and activities Veridian has scheduled for today. Completing them keeps your memory strong — skipping them lets things slip."

Flashcard Session — Rating Buttons (first flashcard review):

"Rate how well you remembered each card honestly. Again means you forgot, Easy means it was effortless. Veridian uses these ratings to decide when to show you each card again."

Practice Quiz — Soft Intervention (first time triggered):

"You've missed a few questions on the same concept. This happens — here's a quick reminder before continuing."

Session Summary (first session completed):

"This summary updates your Journey's strength and weakness map. Check it after each session to see where you're improving and where to focus next."

Journey Detail — Module Stages (first time viewing a module):

"Each module has three stages: Learn, Practice, and Mastery. Veridian moves you through them based on your performance, not a fixed schedule."

Community Library (first visit):

"These are Journeys built by other Veridian students. Clone any Journey to copy it into your account — your progress and schedule will always be your own."

Diagnostic (first Journey, just before it starts):

"This short diagnostic places you in the right starting stage for each module. It only takes a few minutes and means you won't waste time reviewing things you already know."

Learning Profile (first visit with data):

"Your learning profile builds over time as you study. The knowledge map shows everything you've covered and how well each concept is holding."

Phase 6 — Empty States

Several pages in Veridian have meaningful empty states — pages that have no data yet because the student is new. These must be designed intentionally. An empty state that looks broken or blank is a silent churn driver.

Journey List — No Journeys

What to show:

A centered card with:

- Headline: "No Journeys yet"
- Sub-text: "Create your first Journey by uploading study material — notes, a textbook chapter, or a PDF. Veridian will build a personalized study plan from it."
- **Create Journey** button — primary, prominent
- **Browse Community Library** — secondary link below the button

Home Screen — No Journeys, No Due Cards

What to show:

The standard home screen layout but the main content area shows:

- "You're all set up. Build your first Journey to get started."
- **Create Journey** button
- A faint preview of what the home screen looks like when populated — ghosted/skeleton cards that hint at what will appear — this sets expectations without misleading

Due Today — Nothing Due

What to show:

A positive, calm confirmation:

- "You're all caught up."
- Sub-text: "Nothing is due today. Come back tomorrow or explore a new module in one of your Journeys."

This should never feel like an error or an empty screen. Being caught up is an accomplishment and the UI should treat it as one.

Learning Profile — No Study History

What to show:

The Learning Profile tab renders normally but in place of charts and data:

- "Your learning profile will build as you study."
- Sub-text: "Start a session in any Journey to see your first insights here."
- The knowledge map section shows a greyed-out placeholder that hints at what will appear

Community Library — First Visit Before Any Journeys Are Cloned

The Community Library should never have an empty state because it shows other students' Journeys, not the student's own data. But on the first visit, a one-line contextual banner appears at the top of the library — not a hint tooltip, just a soft informational line:

"Browse Journeys created by other Veridian students. Clone any Journey to add it to your account."

This banner is shown once and dismissed automatically after the student browses for more than 30 seconds or manually closes it.

Phase 7 — Re-Onboarding Edge Cases

Student Signs Up, Abandons During Welcome Flow

If a student creates an account and drops off mid-welcome-flow — closes the tab, navigates away — their welcome flow progress is saved. On their next sign-in they resume from where they left off. They do not start over from the beginning.

Student Completes Sign-Up but Never Creates a Journey

A student who finishes the welcome flow by skipping every step has an account but no Journeys. Every session they return to, the home screen shows the empty Journey state with the create prompt. There is no secondary onboarding email sequence in v1, but this is a logical future addition.

Student Clones Their First Journey Instead of Building One

The clone flow from onboarding is identical to the standard Community Library clone flow described in Section 3. The only difference is that after cloning, the student is returned to the home screen rather than staying in the library. The onboarding context is complete once a Journey exists in their account regardless of how it got there.

Design Principles for the Onboarding Experience

Fast over thorough. Every extra second in the onboarding flow is a second where the student might leave. The welcome sequence should take under 90 seconds for a student who engages with all four steps.

Contextual over generic. Every prompt, headline, and call to action should reference what the student actually told Veridian about themselves. A student studying for the MCAT should not see the same copy as a student studying for AP History.

Real over illustrative. Empty states and onboarding prompts should show real product UI wherever possible — a real Journey card, a real module list, a real session summary. Abstract illustrations of studying are forgettable. Seeing the actual product creates desire.

Helpful over impressive. The onboarding flow should not try to show off every feature. Its only job is to get the student to their first real study session. Every element that does not serve that goal should be cut.

Security

Overview

Veridian's infrastructure security — authentication, session management, database access control, hosting, and encryption in transit — is handled entirely by Base44. This appendix does not re-document what Base44 owns. It documents the product-level data boundaries that every Veridian developer must understand and enforce when building any feature. These are the decisions that belong to the application layer, not the platform layer.

When in doubt about whether a piece of data should be visible, accessible, or sent anywhere, this appendix is the reference.

1. User Data Ownership

Every piece of data Veridian stores belongs to the user who created it. This is not just a legal principle — it is a design principle that should be enforced at the query and component level.

No user should ever be able to read, access, or infer another user's private data through any feature, API call, or URL manipulation. Base44's entity access rules enforce this at the database level, but developers must not create UI paths or API patterns that bypass this by accident.

2. Public vs Private Data Boundaries

The following table defines exactly what is and is not visible outside a user's private account.

Journey Data

Data Field	Private (owner only)	Public (when Journey is published)
Module names and descriptions	—	✓ Visible
Activity types and item counts	—	✓ Visible
Tags	—	✓ Visible
Clone count	—	✓ Visible
Rating summary	—	✓ Visible

Creator username	—	✓ Visible
Published date	—	✓ Visible
Source material content	✓ Hidden by default	Creator may enable
Actual question and card content	—	✓ Visible after cloning only
Personal exam deadline	✓ Always private	Never
Diagnostic results	✓ Always private	Never
Module stage placement	✓ Always private	Never
Mastery scores	✓ Always private	Never
FSRS card state	✓ Always private	Never
Session history	✓ Always private	Never
Personal notes or annotations	✓ Always private	Never

Publishing a Journey exposes its structure and content. It never exposes the owner's personal learning data under any circumstances.

User Account Data

Data Field	Visibility
Display name	Private — never shown externally in v1
Username	Public only when attributed to a published Journey in the Community Library
Email address	Always private — never exposed anywhere in the product
Authentication provider	Always private
Notification preferences	Always private
Avatar	Private in v1 — no public profile exists
Joined date	Always private

3. Route-Level Access Control

Every route in Veridian must enforce the correct access level. This is implemented via React Router auth guards.

Route	Access Level
/	Public — unauthenticated allowed
/signup	Public — unauthenticated only, redirect to /home if already signed in
/signin	Public — unauthenticated only, redirect to /home if already signed in
/home	Authenticated only
/journeys	Authenticated only
/journeys/:id	Authenticated only — owner only
/journeys/:id/modules/:moduleId	Authenticated only — owner only
/study/:sessionId	Authenticated only — owner only
/library	Public — unauthenticated allowed
/library/:journeyId	Public — unauthenticated allowed, published Journeys only
/profile	Authenticated only

The `/library/:journeyId` route must validate that the Journey being accessed has `isPublic: true` before rendering. If a user attempts to access a private Journey's preview URL directly, they must receive a 404 or access denied state — not the Journey content, and not an authentication redirect that leaks the existence of the Journey.

Authenticated routes that belong to a specific user — Journey Detail, Module Detail, study sessions — must verify that the requesting user is the owner of that resource. Accessing

another user's Journey URL should return a 404, not an access denied message that confirms the Journey exists.

4. Gemini API Data Rules

The Gemini API receives student data in order to generate content and grade responses. The following rules govern what may and may not be sent.

What May Be Sent to Gemini

- Source material content uploaded by the student for the purpose of generating their Journey
- The student's typed or spoken Feynman Mode explanation for grading
- The student's Free Recall response for grading against the knowledge map
- Session performance data for post-session pattern analysis — accuracy percentages, concept labels, time data. No personally identifying information
- Question text and answer choices for follow-up question generation

What Must Never Be Sent to Gemini

- The student's name, email address, or any account identifier
- The student's exam deadline or any calendar data
- The student's diagnostic results or stage placement
- Raw FSRS card state data
- Any data from one student's account when generating content for another student's account
- Any content the student has not explicitly uploaded or submitted for AI processing in that specific session

Prompt Engineering Security

AI prompts must not be constructed in a way that allows student-supplied input to alter the system prompt or change the behavior of the AI call. This is a prompt injection risk. Any student-supplied text — uploaded documents, typed explanations, pasted notes — must be passed as data content in the prompt, clearly delimited and separated from the system instruction portion of the prompt.

5. File Upload Boundaries

Students upload PDF and document files as source material for Journey creation. The following rules apply:

- Uploaded files are processed by the Gemini Files API for content extraction and then used to generate Journey content
 - Files should not be stored indefinitely beyond what is needed for Journey generation and display. Base44 handles storage but the application should not retain raw uploaded files longer than necessary
 - File uploads must be limited to document types only — PDF, plain text, common document formats. Image-only files without extractable text should be rejected with a clear error message
 - Maximum file size should be enforced at the UI level before upload begins — a reasonable limit for v1 is 20MB per file. This prevents abuse and controls Gemini Files API costs
 - Students may not upload files on behalf of other users. File uploads are always scoped to the uploading user's own Journey creation flow
-

6. Community Library Abuse Prevention

The Community Library is publicly accessible and user-generated. Basic safeguards must be in place from day one:

- Published Journeys must meet minimum content requirements before they are visible in the library — at least 3 modules, at least one complete activity per module, at least one tag. This is enforced at publish time and documented in Section 3
 - A Journey that is unpublished by its creator is removed from the library immediately and is no longer accessible via its public URL. Clones already made by other users are unaffected
 - The rating and tagging system on Journey previews is limited to users who have cloned and actively used the Journey for at least 3 days. This prevents rating manipulation by the creator or their associates
 - In v1 there is no automated content moderation. If abuse patterns emerge — spam Journeys, inappropriate content — the manual response is to unpublish via the Base44 admin panel. A reporting mechanism is a v2 addition
-

7. Hint and Onboarding State

The hint system tracks which onboarding hints a user has seen to prevent them from repeating. This data is stored on the user entity in Base44 as an array of seen hint IDs.

This data is:

- Private to the user
 - Never used for analytics or behavioral profiling in v1
 - Never shared with third parties
 - Deletable as part of account deletion
-

8. Account Deletion

When a student deletes their account, the following must be permanently removed:

- All Journeys owned by the student — including all modules, activities, cards, and session history
- All FSRs card state and scheduling data
- All Learning Profile data — mastery scores, session history, subject trends
- Account credentials and personal information
- Hint and onboarding state
- Any cached or derived data associated with the user ID

What is not deleted:

- Clones of the student's published Journeys that other users have already made. These are independent copies owned by those users and are not affected by the original creator's account deletion
- The student's username attribution on Community Library Journey cards may be anonymized rather than deleted — shown as "Deleted User" — to preserve the integrity of the library for students who cloned those Journeys

Account deletion is permanent and irreversible. The double confirmation modal described in Section 4 must make this clear before the action is taken.

9. Summary of Developer Rules

These are the non-negotiable rules every developer must follow when building any Veridian feature:

- Never expose private user data through public routes, shared URLs, or API responses not scoped to the requesting user

- Never send personally identifying information to the Gemini API
- Never allow a URL parameter to return another user's private resource — return 404, not 403, to avoid confirming existence
- Always validate that a published Journey has `isPublic: true` before rendering it on a public route
- Always enforce file type and size limits before upload begins
- Always scope Base44 entity queries to the authenticated user's ID — never fetch all records of a type without a user filter
- Always remove all user data on account deletion — no orphaned records

Finance Log

Date	Item	Cost
	NameCheap Domain	

Parking Lot

